

## Duan Gao (高端)

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### Education

- [2017.9 ~ 2022.6] Tsinghua University, Ph.D., Computer Science  
Advised by Prof. Kun Xu  
Research interests: Computer Graphics (Appearance modeling, physically-based rendering and neural rendering)
- [2012.9 ~ 2017.6] Nanjing University, B.A., Computer Science  
Advised by Prof. Jie Guo  
Research interests: Computer Graphics
  - 2012.9 ~ 2014.9, Nanjing University, Life Science
  - 2014.9 ~ 2017.6, Nanjing University, Computer Science

### Experience

- 3D Generation Algorithm Engineer @X Square Robot, 2025 ~ *Present*  
3D generation algorithm engineering and system development for embodied robotics applications.
- Principal Engineer, Top Minds(天才少年) @Huawei Cloud, Media Innovation Lab, 2022 ~ 2025  
Generative 3D digital humans, Diffusion-based Physically-based material generation, AIGC-3D, Differentiable rendering.
- Research intern @Microsoft Research Asia, Internet Graphics Group, 2019 ~ 2020  
Neural relighting research with Dr. Guojun Chen, Dr. Yue Dong and Dr. Xin Tong.
- Research intern @Tencent AI Lab, 2019.7 ~ 2019.8  
3D face related research with Dr. Haozhi Huang.
- Research intern @Microsoft Research Asia, Internet Graphics Group, 2018 ~ 2019  
Appearance modeling research with Dr. Yue Dong and Dr. Xin Tong.
- Research intern @Megvii(Face++), 2017.7 ~ 2017.9

Facial recovery and rendering research with Dr. Liqian Ma.

## Selected Awards

### 2022

- Huawei Top Minds Program(华为天才少年)
- Outstanding Graduate of Beijing
- Outstanding Doctoral Graduate, Department of computer science and technology, Tsinghua University

### 2021

- Style3D Graduate Fellowship (10 winners annually)
- First Prize Award for Presentation at the Inaugural Journal of Image and Graphics (JIG) Graduate Academic Forum
- Fall Comprehensive Scholarship, Tsinghua University

### 2019 and before

- “Star of Tomorrow” Intern Award, Microsoft Research Asia, 2019
- National Encouragement Scholarship, 2015
- National Encouragement Scholarship, 2013

## Publications

1. ControlFace: Few-Shot 3D Face Generation via a Controllable Diffusion Model Guided by Text and Images  
*IEEE International Conference on Multimedia & Expo (ICME), 2025 (CCF B)*  
Jinfu Wei, Zheng Zhang, Qinchuan Zhang, Ran Liao and Duan Gao
2. DreamPBR: Text-driven Generation of High-resolution SVBRDF with Multi-modal Guidance  
*IEEE International Conference on Multimedia & Expo (ICME), 2025 (CCF B)*  
Linxuan Xin, Zheng Zhang, Zhiyi Pan, Jinfu Wei, Duan Gao, Wei Gao
3. UniFaceGAN: High-Quality 3D face Editing with a Unified Latent Space  
*IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2025 (CCF B)*  
Jinfu Wei, Zheng Zhang, Ran Liao and Duan Gao

4. Neural Global Illumination: Interactive Indirect Illumination Prediction under Dynamic Area Lights  
*IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2022 (**CCF A**)  
**Duan Gao**, Haoyuan Mu, Kun Xu
5. Deferred Neural Lighting: Free-viewpoint Relighting from Unstructured Photographs  
*ACM Transactions on Graphics (Proceedings of SIGGRAPH Asia 2020)* (**CCF A**)  
**Duan Gao**, Guojun Chen, Yue Dong, Pieter Peers, Kun Xu, Xin Tong
6. Deep Inverse Rendering for High-resolution SVBRDF Estimation from an Arbitrary Number of Images  
*ACM Transactions on Graphics (Proceedings of SIGGRAPH 2019)* (**CCF A**)  
**Duan Gao**, Xiao Li, Yue Dong, Pieter Peers, Kun Xu, Xin Tong
7. High-frequency Face Estimation from a Single Image  
*CCF CAD&CG*, 2019  
**Duan Gao**, Kun Xu

#### Patents

1. A material map generation method and device and computing device  
 CN120726206A, 2025.09.30  
 Duan Gao, Linxuan Xin, Zheng Zhang
2. A method, system and related equipment for adjusting three-dimensional model  
 CN120219679A, 2025.06.28  
**Duan Gao**, Zheng Zhang, Yuan Gao, Jinfu Wei
3. Data processing method and data processing device  
 CN119963699A, 2025.05.09  
**Duan Gao**, Zheng Zhang
4. Rendering method and device  
 CN116266376A, 2023.06.30  
**Duan Gao**, Kun Xu, Chengyu Gu, Jingtang Liao
5. Three-dimensional face model construction method and device, computer equipment and storage medium  
 CN110675413B, 2020-11-13  
**Duan Gao**, Shilong Zhou, Yonggen Ling, Haozhi Huang, Shi-Min Hu, Kun Xu, Wei Liu

6. Face image data acquisition method and face image data acquisition device

*CN108876891B, 2021-12-28*

Liqian Ma, **Duan Gao**

7. Face image synthesis method and device

*CN107644455B, 2022-02-22*

Liqian Ma, **Duan Gao**

#### **Academic Services (review)**

- CCF-A: SIGGRAPH 2022/2025, SIGGRAPH Asia 2023/2025
- CCF-B: Eurographics(EG), Pacific Graphics(PG), The Visual Computer, Journal of Computer-Aided Design & Computer Graphics (JCAD), etc.